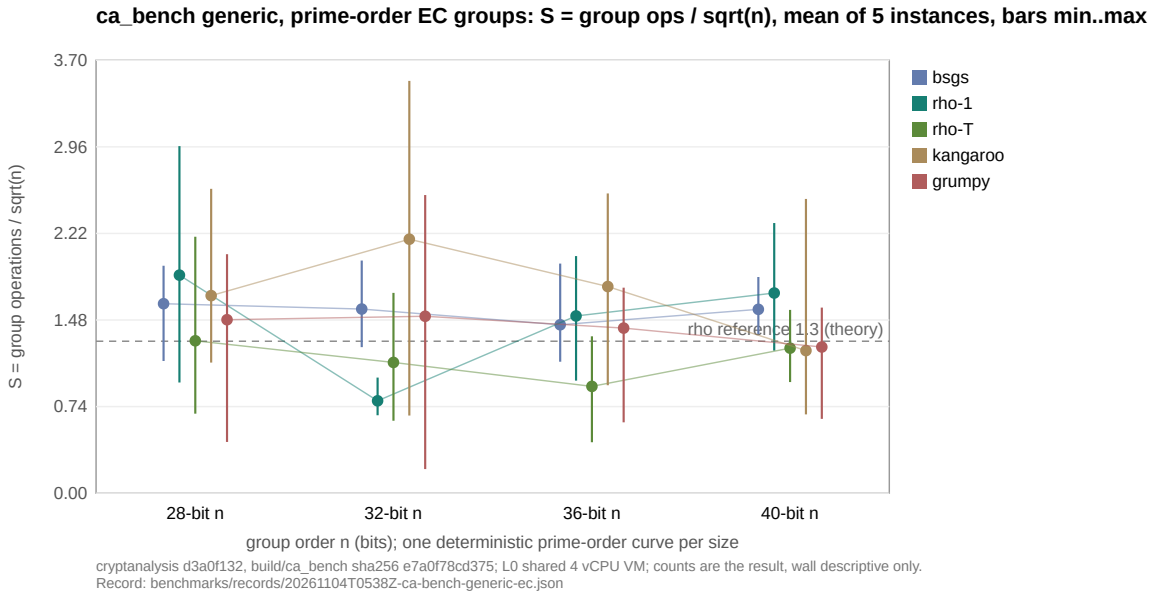
ca_bench generic on prime-order EC groups: the first benchmark record

Date. 2026-11-04. **Status.** Measured `[pcc:unverified]`: counts are read from the harness's own `ca_stats` and no independent verifier has checked them. **Class.** Engineering measurement, no research claim: this run seeds the benchmark progress report (`docs/benchmark-records.md`) with its first `bench.record/v1` point in the headline unit `S`. **Record.** `benchmarks/records/20261104T0538Z-ca-bench-generic-ec.json`. **Host.** Shared 4 vCPU cloud VM, no CPU pinning, no hardware counters: **L0 in this repository's tiers**. Wall time is descriptive only; the reported quantity is the group-operation count, which the solver counts itself.



Question

What does the native generic-DLP suite (`build/ca_bench generic`) cost today on prime-order elliptic-curve groups at 28, 32, 36 and 40 bits, in `S = group operations / sqrt(n)`, so that a later run of the same command has a recorded first point to be compared with?

Method

- Commit `d3a0f1320886` (main), `cmake -DCMAKE_BUILD_TYPE=Release`, target `ca_bench`; binary SHA-256 `e7a0f78cd3754cd524a4e89ca5bc9148ec37192dfeb957d33d60c0c38ab9bebf`.
- Command, run under the record with `tools/benchrec.py run`: `build/ca_bench generic --group ec --bits 28,32,36,40 --reps 5 --threads 4 --json experiments/ca-bench-generic-ec-20261104/raw.jsonl`
- Instances: `ca_bench`'s deterministic curve per bit size (`y^2 = x^3 + x + b` over the first prime above `2^(bits-1)|5`, smallest `b` giving prime order), 5 random targets each, every one solved (`ok` 5/5 in every cell).
- Arms: `bsgs`, `rho-1` (one thread), `rho-T` (4 threads), `kangaroo`, `grumpy`. The matched reference for the record's `value/ref` column is this run's own `rho-1` arm on the same

40-bit curve, so the reference shares instance, build and host. It is not the frozen rho reference of the IC leaderboard and does not enter the §5a band.
- Raw: `raw.jsonl` (JSON Lines: header, then one row per solved instance with the `ca_stats` counters), `stdout.txt`, `summary.json` (per-cell means, min, max). Each is in the content-addressed store by SHA-256; the record names the S3 URIs.

bits	p	a	b	n
28	134217757	1	70	134234689
32	2147483659	1	230	2147540641
36	34359738421	1	77	34359573251
40	549755813911	1	91	549756147371

Results (measured)

S per cell: mean over 5 instances, with the minimum and maximum instance.

bits	algorithm	S mean	S min	S max	ok	wall s (5 instances)
28	bsgs	1.621	1.131	1.946	5/5	0.0054
28	rho-1	1.866	0.947	2.970	5/5	0.0074
28	rho-T	1.305	0.680	2.194	5/5	0.0067
28	kangaroo	1.690	1.118	2.604	5/5	0.0063
28	grumpy	1.485	0.438	2.046	5/5	0.0164
32	bsgs	1.575	1.249	1.990	5/5	0.0247
32	rho-1	0.791	0.667	0.990	5/5	0.0123
32	rho-T	1.118	0.619	1.714	5/5	0.0163
32	kangaroo	2.174	0.664	3.528	5/5	0.0194
32	grumpy	1.514	0.206	2.552	5/5	0.0884
36	bsgs	1.441	1.125	1.964	5/5	0.0991
36	rho-1	1.517	0.963	2.029	5/5	0.0945
36	rho-T	0.913	0.435	1.342	5/5	0.0505
36	kangaroo	1.768	0.924	2.565	5/5	0.0650
36	grumpy	1.413	0.606	1.758	5/5	0.3485
40	bsgs	1.573	1.361	1.850	5/5	0.5717
40	rho-1	1.713	1.220	2.312	5/5	0.3594
40	rho-T	1.241	0.951	1.568	5/5	0.2857
40	kangaroo	1.220	0.674	2.518	5/5	0.1413
40	grumpy	1.251	0.636	1.588	5/5	1.7549

Fitted exponents over the four sizes (from the harness): bsgs 0.493, rho-1 0.514, rho-T 0.487, kangaroo 0.457, grumpy 0.479 against the theoretical 0.5; R^2 0.96 to 0.9994. Five instances per cell is too few for a rho constant to mean much: rho-1 at 32 bits came in at 0.79 and at 36 bits at 1.52, which is the spread of five random walks, not a change in the algorithm.

Distance to the matched reference at 40 bits (derived: value / rho-1 mean S = 1.713):

algorithm	S mean	value / reference
bsgs	1.573	0.918
rho-T	1.241	0.725
kangaroo	1.220	0.712
grumpy	1.251	0.731

What this does and does not establish

- It establishes a recorded first point per arm at 40 bits in `S`, with raw rows in S3 and the command, build and host in Git. The next run of the same command yields the progress report's first gain-vs-previous entry.
- It does not establish a speedup, a ranking between arms, or anything about the index-calculus baseline: no code changed, n = 5 per cell, L0 host.
- Limits: wall time on a shared VM; the reference is the run's own rho arm, not a frozen external reference; no replay certificate.

Evidence

- Record: `benchmarks/records/20261104T0538Z-ca-bench-generic-ec.json`
- Progress report regenerated: `benchmarks/progress/` (`PROGRESS.md`, `progress.svg`, `progress.pdf`, `progress.json`)
- Benchmark entry: `benchmarks/20261104-ca-bench-generic-ec/`
- Graph source: this report's generator is the inline Python in the session that produced `figures/s_by_bits.svg` from `summary.json`; `summary.json` is the editable data.